<table>
<tr><td>Story Name</td><td>Controller</td><td>Duration</td><td>2 weeks</td></tr>
<tr><td rowspan="3">Informal description</td><td colspan="3">As an employee of EcoWeighting S.A.</td></tr>
<tr><td colspan="3">... I would like a functioning weighting controller</td></tr>
<tr><td colspan="3">... so I can use my QR-code to weight and dispose my waste.</td></tr>
<tr><td>Requirements according to the work order</td><td colspan="3">

Must

All the necessary libraries have been installed.
The ESP32 is flashed with firmware.
The firmware can be executed without errors, and it correctly controls the individual hardware elements.
The firmware was adapted for the integration of the webserver.

Connection with WIFI
Specification of server name and file to access
ID of balance

The documentation contains the relevant steps for flashing and setting up the scale.

Should

Set up the project as an independent scale (no QR-code reader or webserver integration).
The scale is calibrated by setting the two parameters based on the calculation of a linear equation.

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<tr><td colspan="4">Tasks according to the work order</td></tr>
<tr><td colspan="4">Install the Arduino Ide 2.1.0 (Download link: https://www.arduino.cc/en/software#future-version-of-the-arduino-ide .)</td></tr>
<tr><td colspan="4">Download and prepare the QR Reader Library using the following tutorial
https://www.makeuseof.com/scanning-qr-codes-with-esp32-cam/</td></tr>
<tr><td colspan="4">Install the libraries in the Arduino ide using the LIBRARY MANAGER:

esp32 (by Expressif Systems)
Adafruit SSD 1306 (by Adafruit)
ESP8266 and ESP32 OLED driver for SSD1306 displays (by ThinkPulse)
HX711 Arduino Library (by Bogdan)

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- PubSubClient (by Nick O'Leary)
- TaskScheduler (by Anatoli Arkhipenko)
- Bounce2 (by Thomas O Fredericks)

Use the following tutorial: https://techtutorialsx.com/2017/12/02/esp32-arduino-interacting-with-a-ssd1306-oled-display/ to create a script which can display a text on the OLED. Use "Hello World" as an example text.

!Important! Use the pins **13** and **15** instead of 21 and 22.

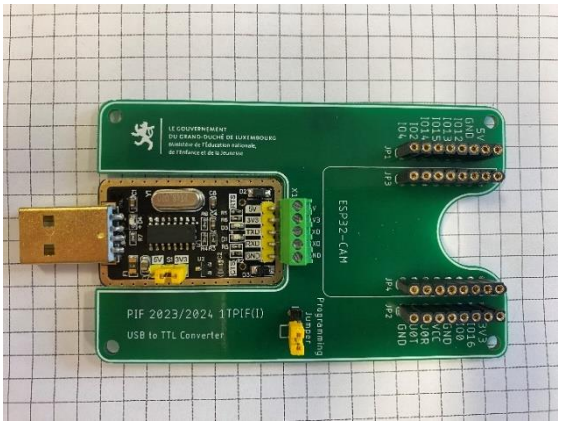
→ `SSD1306 display(0x3c, 13, 15);`

Hint: Use the function flipScreenVertically() to change the orientation of the text.

Use the following tutorial https://randomnerdtutorials.com/esp32-load-cell-hx711/

Go to the section **Calibrating the Scale (ESP32 with Load Cell)** to create a script which can take a measurement using the scale.
!Important! Use the pins **HX711_SCK 14** and **HX711_DOUT 2**
Change this script to output the result (measured weight) on the OLED display instead of using the serial port.

Flash the ESP32 CAM with the script. Use the USB to TTL Converter and select the correct board and port.



Calibrate the scale:

1. Call `set_scale()` with no parameters.
2. Call `tare()` with no parameters
3. Place a known weight on the scale and call `get_units(10)`
4. Divide the result in **step 3** to your known weight. You should get about the parameter you need to pass to `set_scale()`.
5. Adjust the parameter in **step 4** until you get an accurate reading.

Find and adapt the following parameters in the provided code:

- the SSID,
- the password of the WLAN,
- the host or domain name of the server,
- and the file to access.

Create the documentation.